PUBLICATION

Automated Charge Capture: How AI Documentation Closes the Revenue Gap

How ClinixSummary identifies billable procedures, supplies, and interventions from clinical encounters — reducing charge leakage and improving revenue integrity. Pilot data shows 30% improvement in charge capture.

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1. Executive Summary

Charge capture — the process of identifying and recording all billable services, procedures, supplies, and equipment used during a patient encounter — is one of healthcare's most persistent revenue challenges. The Healthcare Financial Management Association (HFMA) estimates that hospitals lose approximately 1% of net revenue to charge capture leakage alone, with broader billing and documentation failures costing 3–5% of total revenue.

Sources: HFMA, "Avoiding the High Cost of Charge-Capture Leakage"; Advisory Board, Revenue Cycle Benchmarks, 2023.

ClinixSummary's automated charge capture module addresses this gap at the source: by identifying every billable item — procedures, interventions, equipment, supplies — directly from the clinical encounter audio, matched against the organisation's own charge master. Pilot deployment data demonstrates a 30% improvement in charge capture within the first month.

2. The Charge Capture Problem

Manual charge capture is inherently error-prone. Clinicians are focused on patient care, not billing — and documenting every supply used, every piece of equipment deployed, and every procedure performed is a secondary concern during the encounter. The result is systematic under-capture of billable items:

- Supplies and equipment — items used during procedures (suture kits, splints, dressings, injectable medications) are frequently omitted from charge records.
- Ancillary procedures — minor procedures performed during a consultation (wound irrigation, joint injection, nebuliser treatment) may go unbilled if the clinician forgets to add them.
- Level-of-service under-coding — documentation that lacks specificity leads to conservative coding, resulting in lower reimbursement than the encounter warrants.
- Time-based services — prolonged service codes, critical care time, and counselling time are frequently under-reported because clinicians do not track time precisely.

For a hospital with $500 million in net patient revenue, even 1% charge leakage represents $5 million in annual lost revenue — revenue that was earned through care delivery but never billed.

3. Evidence: Scribes and Charge Capture

The positive impact of scribes on charge capture and revenue is well-documented in peer-reviewed literature:

- A systematic review and meta-analysis by Gottlieb et al. (Annals of Emergency Medicine, 2021) covering 39 studies and over 562,000 encounters found that scribes increased RVUs per encounter by 0.14 and RVUs per hour by 0.55 — reflecting more complete documentation and billing.
- Bank et al. (ClinicoEconomics and Outcomes Research, 2013) demonstrated a 57% increase in work RVUs per hour in a cardiology practice after scribe deployment, with a follow-up study (2015) showing $1.37 million in additional annual revenue at a scribe cost of approximately $99,000.
- Heaton et al. (Journal of Emergency Medicine, 2017) found a positive impact on billed RVUs for adult patients, particularly in higher-acuity encounters.

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Sources: Gottlieb M et al., Ann Emerg Med, 2021;77(2):180-189; Bank AJ et al., ClinicoEcon Outcomes Res, 2013;5:399-406; Heaton HA et al., J Emerg Med, 2017;52(3):370-376.

These studies demonstrate that more complete documentation — whether produced by human scribes or AI — directly translates to more accurate charge capture and higher revenue. ClinixSummary extends this principle beyond documentation into active charge identification.

4. ClinixSummary's Charge Capture Approach

ClinixSummary goes beyond passive documentation improvement. Our automated charge capture module actively identifies billable items from the clinical encounter and maps them to the organisation's billing codes:

4.1 Charge Master Integration

Organisations provide their charge master — the comprehensive list of procedures, supplies, equipment, and services with associated billing codes (CPT, HCPCS, internal codes). ClinixSummary ingests this charge master and uses it as the reference for charge identification.

4.2 Encounter-Based Charge Detection

During the clinical encounter, ClinixSummary's NLP engine listens for billable events:

- Procedures performed — "I'm going to inject the left knee with 40mg of triamcinolone" triggers identification of joint injection (CPT 20610) plus the injectable medication.
- Supplies used — "let's use a 4-0 Vicryl for the deep layer and 5-0 Prolene for skin" captures suture supplies and the associated repair code.
- Equipment deployed — "I'll set up the nebuliser with albuterol" identifies nebuliser treatment (CPT 94640) and the medication.
- Time-based services — encounter duration is tracked automatically, supporting appropriate E/M level coding and prolonged service add-on codes.
- Diagnostic testing — point-of-care tests, EKGs, and imaging orders mentioned during the encounter are captured for charge reconciliation.

4.3 Charge Reconciliation Output

At the conclusion of each encounter, ClinixSummary generates a charge capture summary: a list of identified billable items mapped to the organisation's charge master codes, with supporting clinical documentation for each charge. This summary is presented alongside the clinical note for clinician review and approval before submission to billing.

5. Pilot Results: 30% Charge Capture Improvement

ClinixSummary's automated charge capture module was deployed in a one-month pilot across multiple clinical settings. Results compared charge capture rates before and after deployment:

- 30% increase in identified billable items per encounter compared to pre-deployment baseline.
- Supply charges were the most impacted category, with a 42% increase in captured supply items.

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- Ancillary procedure charges increased by 28%, reflecting procedures that were previously performed but not billed.
- E/M level accuracy improved, with a 19% reduction in under-coded encounters.
- Average charge lag (time between service and charge entry) decreased from 3.2 days to same-day.

Clinician feedback was overwhelmingly positive: 94% reported that the charge capture summary was accurate and useful, and 89% said it identified charges they would otherwise have missed.

6. Implementation

Deploying ClinixSummary's charge capture module requires three steps:

- Charge master upload — the organisation provides its current charge master in standard format. ClinixSummary's integration team maps it to the detection engine.
- Calibration period — a 1–2 week period where the system learns the organisation's specific billing patterns, commonly used supplies, and procedure preferences.
- Go-live — charge capture summaries are generated for every documented encounter, reviewed by clinicians, and integrated into the existing billing workflow.

The module integrates with existing practice management and billing systems via FHIR-compatible APIs, ensuring no disruption to current revenue cycle workflows.

7. Conclusion

Charge capture leakage is a solved problem. When AI documentation systems actively identify billable items from the clinical encounter and match them to the organisation's charge master, the revenue gap closes. ClinixSummary's automated charge capture module delivers measurable financial impact from day one — capturing revenue that was earned through patient care but previously lost to manual processes.

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